Jonah Cha

ML research and engineering: evaluation, RL, robot learning, foundation models

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Education

B.S. Electrical Engineering and Computer Science, UC Berkeley · Expected May 2028 · GPA 4.00 · Berkeley, CA

Research

CyberGym-E2E: Scalable Real-World Benchmark for AI Agents' End-to-End Cybersecurity Capabilities ICML 2026

T. Shi*, R. Rheem* (* equal contribution), ..., **J. Cha** (10th of 16), ..., D. Song · arxiv.org/abs/2606.04460

Proceedings of the 43rd International Conference on Machine Learning, PMLR 306, pp. 112094–112108

Counterfactual Action Editing with a World Model Jul–Aug 2026 · Robot learning, simulation

Negative result · Built with Claude Code: A probe on the frozen encoder's features finds where a robot episode went wrong (AUROC 0.88–0.92; a position-only control scores 0.42–0.48). But in eight deployable configurations, no searched edit beat a random edit of the same size. github.com/MERC13/failure-detection-WM

Can a frozen VLA be steered through its own instructions? Jul 2026 · Robot learning, simulation

Negative result · Built with Claude Code: An audit before any RL: rephrasing the instruction given to a frozen π0.5 barely moves its success on LIBERO. Only 2 of 8 tasks showed headroom for a learned steering policy, and the base checkpoint completed 0 of 30 episodes. github.com/MERC13/openpi/tree/steering-rl/steering_rl

Experience

University of California, Berkeley Researcher Aug 2026 – present

Foundation models for RNA encoding and embedding, especially sequence-to-structure prediction: new transformer architectures and representation-learning objectives, with compute optimisation on Trainium.

University of California, San Francisco Researcher, two projects Jan 2026 – present

Diffusion transformer models that reconstruct video from fMRI, benchmarked against published fMRI-to-video methods (MinD-Video, NeuroClips) on the field's standard semantic and pixel-level tests. Separately, deep learning and quantitative imaging tools for early detection and risk stratification in clinical radiology.

Mercor Intelligence Machine Learning Engineer Sep 2025 – Aug 2026

Post-training for AI agents on stealth projects, especially reinforcement learning for agents: training examples, and evaluations of agent behaviour.

Berkeley AI Research Researcher Sep – Dec 2025

CyberGym-E2E with Dawn Song's group at Berkeley RDI; the publication is above.

Stoa USA Software Engineering Intern Jun 2024 – Aug 2025

Built StoaHub, the national web platform of the Stoa speech and debate league, including the rankings page that tabulates results across its tournaments.

Uplevel Ops Legal Consulting AI Consultant May 2024 – Aug 2025

Cloud AI agents for a legal consultancy's clients: NDA drafting from a clause library, contract redlining against a playbook, and invoice review against billing guidelines. Uplevel estimated over $2 million in client value.

About

Coursework: Machine Learning (A+), Data Structures (A+), Computer Programs (A), Discrete Mathematics and Probability Theory (A), Reinforcement Learning (audited), Deep Learning (audited, in progress), Computer Vision (in progress), Computer Architecture (in progress)

Skills

Languages: Python, C++, Java, PHP, JavaScript, SQL, HTML and CSS
ML and infrastructure: PyTorch, HuggingFace, JAX (through openpi), Docker, AWS including Trainium, Linux, Git, Redis, Pinecone, Laravel
Agents: LangChain, LangGraph, LangSmith, AutoGen, MCP, Claude

Awards: SAT 1600/1600, PSAT 1520/1520, AP Scholar with Distinction, 2× National Gold Medal Debater, Presidential Scholar Candidate.